

CERTICUT Layered Workflow

Quantum circuit + fixed representation
independent-QPD gate costs $\rho_g = (\sum_i |a_{g,i}|)^2$

Core independent-QPD reduction
 $w_{ij} = \sum_{g \in G_{ij}} \log \rho_g$
capacitated K -way graph partitioning

$K = 2$ reference
B2S + H2 + B&B
structural ablation

Production backend
SCIP K -way MIP
solver-tolerance
primal/dual bounds

Solver-tolerance model-overhead factor
 $\text{LB} \leq \text{OPT} \leq \text{UB} \quad F = \exp(\text{UB} - \text{LB})$